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| COMP 2121 DISCRETE MATHEMATICS | LAB 10 |
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3. Build Huffman tree based on “send beans and banana seeds” and provide codeword for each character. What is the total number of bits required to encode the sentence.

4. A certain file contains only two characters, A and B:

Char	Normalized Frequency
A	0.7
B	0.3

According to Shannon's Formula the lower bound on the number of bits per character is:

$$\sum_i^n p_i \log_2 \left(\frac{1}{p_i} \right) = 0.7 \cdot \log_2 \left(\frac{1}{0.7} \right) + 0.3 \cdot \log_2 \left(\frac{1}{0.3} \right) = 0.88129$$

- a) Build Huffman tree and calculate the expected number of bits per character. Compare this result with the lower bound.

- b) Based on the original information calculate the normalized frequency of symbols composed of two characters.

Build Huffman tree and calculate the expected number of bits per symbol and expected number of bits per character. Compare this result with the lower bound.

Example of Recursive Algorithm and Mathematical Induction.

Example of 4x4 board with one square missing, covered with L-shapes

1	1	2	2
1	3		2
4	3	3	5
4	4	5	5

For any integer $n \geq 1$, if any one square is removed from a $2^n \times 2^n$ checkerboard, the remaining squares can be covered with L-shaped pieces that cover three squares (i.e. trominoes).